Work along a displacement

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Learning target: Calculate work and identify when a force does no work.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 3, Day 13.*
Shift 15 m. Assumes a constant added transverse velocity and a straight-line comparison.

02. Worked example: explain the model
Parallel arrows; rectangle 5 N high and 3 m wide, area 15 J.

03. Read and interpret the model
Both right; positive work.

04. Apply the physics
40 J; 20 J.

05. Explain your reasoning
No displacement so zero mechanical work on bag. Muscles can still use internal chemical energy.

06. Review: connect an earlier idea *Review: Unit 3, Day 4.*
The new force magnitude is 1/4 of the original because the same momentum change occurs over 4 times the time.

07. Exit check
15 J.

08. STRETCH: test the model
Examples 4 N over 5 m and 10 N over 2 m; force parallel to displacement.

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01. Warm-up: bring back an earlier idea *Review: Unit 3, Day 13.*
Shift multiplier 5/3≈1.667 because shift=(J/M) t. Assumes constant added transverse velocity.

02. Worked example: explain the model
Parallel arrows; rectangle 5 N high and 3 m wide, area 15 J.

03. Read and interpret the model
Both right; positive work.

04. Apply the physics
+90 J,-30 J, net +60 J.

05. Explain your reasoning
No displacement so zero mechanical work on bag. Muscles can still use internal chemical energy.

06. Review: connect an earlier idea *Review: Unit 3, Day 4.*
J=-16 N·s; first F≈-10.67 N; longer-time F≈-5.333 N. Both oppose the initial velocity.

07. Exit check
24 N; 0 J.

08. STRETCH: test the model
Trapezoid area (2+8)/2×4=20 J; using final force gives 32 J and overestimates.